

PARALLEL COMPUTER ARCHITECTURE AND PROGRAMMING

[Revised Credit System]

(Effective from the academic year 2018)

SEMESTER - VI

Course Code	CSE 3252	IA Marks	50
Number of Lecture Hours/Week	03	Exam Marks	50
Total Number of Lecture Hours	36	Exam Hours	03
CREDITS – 03			
Course objectives: This course will enable students to • Describe the architecture of GPU, need of GPU computing and different programming languages and models. • Illustrate point-to-point and collective communication primitives in MPI. • Explain the architecture of OpenCL and write efficient kernel functions using OpenCL. • Understand basic programming concepts of CUDA and program parallel patterns like convolution, sparse-matrix multiplication, prefix scan • Implement CUDA programs using shared memory, synchronization of threads, image processing functions			
Module -1			Teaching Hours
INTRODUCTION TO PARALLEL ARCHITECTURES: Introduction to Parallel processing, Parallel Computer Structures: Pipeline Computers, Array Computers, Multiprocessor Systems, Architectural Classification Schemes: Multiplicity of Instruction-Data streams, Introduction, GPUs as parallel computers, Architecture of a modern GPU, Need for parallelism, Parallel programming languages and models Textbook 4: Chapter 1.1.2, 1.3.1-1.3.3, 1.4.1 TextBook 1: 1.1, 1.2, 1.3, 1.4			5 Hours
Module – 2			
MESSAGE PASSING PROGRAMMING: Introduction, Message passing model, MPI basic data types and functions, Point-to-point communication- MPI_Send, MPI_Recv, MPI_Ssend, MPI_Bsend, Collective communication- MPI_Bcast, MPI_Scatter, MPI_Gather, MPI_Reduce, MPI_Allgather, MPI_Alltoall, MPI_Scan, Benchmarking parallel performance, MPI error handling functions. Text Book 2: Chapter 4.1, 4.2, 4.4.1 - 4.4.5, 4.5, 4.6, 6.5			6 Hours
Module – 3			
OpenCL ARCHITECTURE AND PROGRAMMING: Introduction, OpenCL standard, OpenCL specification, Kernels and openCL execution model, Platform and Devices, Execution Environment- Context, Command Queues, Buffers, Program Object and Kernel Object, Program layout, Memory model, Writing Kernels, OpenCL APIs, OpenCL programs for vector-vector addition, Selection Sorting			6 Hours

Text Book 3 : Selected Topics from Chapter 2	
Module – 4	
INTRODUCTION TO CUDA AND COMPUTATIONAL PATTERNS: Introduction, Data Parallelism, CUDA Program Structure, Vector-Vector Addition, Matrix-Matrix Multiplication, Device memories and Data transfer, Kernel functions and Threads, Runtime APIs and Error Handling, 1D Sequential convolution, 1D Parallel Convolution, Atomic and Arithmetic functions, Constant Memory and caching, Tiled 1D Convolution with halo elements, A simple parallel scan algorithm, Sequential Sparse – Matrix vector multiplication, Parallel SPVM using CSR Text Book 1: Chapter 3.1, 3.2, 3.3, 3.4, 3.5, 8.1, 8.2, 8.3, 8.4, 8.5, 9.2, 10.1, 10.2	12 Hours
Module – 5	
CUDA THREADS AND MEMORY ORGANIZATION: Introduction, CUDA Thread Organization, Importance of Memory Access Efficiency, CUDA Device Memory types, synchronization and transparent scalability, strategy for reducing global memory traffic, A tiled matrix-matrix multiplication kernel, parallel image processing applications Text Book 1: Chapter 4.1, 4.4, 5.1, 5.2, 5.3, 5.4 Text Book 5 : Chapter 3.1, 3.2	7 Hours
Course outcomes:	
After studying this course, students will be able to: 1 Understand the concepts related to architecture of GPU, parallel programming languages and models. 2 Write MPI programs using point-to-point and collective communication primitives. 3 Solve parallel programming tasks using OpenCL. 4 Design and implement basic parallel programming concepts using CUDA for different parallel patterns. 5 Implement optimized parallel solutions using thread and memory organization in CUDA	
Text Books: 1 D. Kirk and W. Hwu , “<i>Programming Massively Parallel Processors –A Hands-on approach</i>”, Elsevier Inc.,2nd Edition, 2013. 2 Michael J. Quinn, “<i>Parallel Programming in C with MPI and OpenMP</i>”, McGraw Hill Edition, 2003. 3 Benedict R. Gaster, Lee Howes, David R, Perhaad Mistry, Dana Schaa, “<i>Heterogeneous Computing with OpenCL</i>”, Elsevier Inc., 1st Edition, 2012. 4 Kai Hwang and Faye A. Briggs, <i>Computer Architecture and Parallel Processing</i>, (2e), TMH Private Ltd., 2012. 5 Gonzalez, Rafael C., and Richard E. Woods. "Digital image processing" <i>Publishing house of electronics industry</i>141.7 (2002).	
Reference Books: 1 V.Rajaraman, C. Siva Ram Murthy, “<i>Parallel Computers Architecture and Programming</i>” Prentice-Hall India, 2000. 2 Shane Cook, “<i>CUDA Programming: A developer’s guide to parallel computing with GPUs</i>”, Morgan Kaufman Publication, Elsevier, 2013. 3 Jason Sanders, Edward Kandrot, “<i>CUDA By example: An Introduction to General Purpose GPU Programming</i>”, Addison Wesley, 2011. 4 “<i>CUDA C Programming Guide</i>”, nVIDIA, 2012.	